Stochastic Methods for Finance - Report 4

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Underlying 1: DAX 40 - .GDAXI **Underlying 2:** FTSE MIB - .FTMIB

Every step implementation can be checked in the provided file: `Report4_Quartucci_Davide.xlsx`

0.1 Data Retrieval and Asset Selection

Proper portfolio risk analysis requires the selection of appropriate assets. Two major European indices were selected for this analysis:

- **DAX 40** (`.GDAXI`): Germany's primary stock market index.
- **FTSE MIB** (`.FTMIB`): Italy's main stock exchange index.

Historical data were retrieved via **Refinitiv Workspace**, comprising daily closing prices for both indices over a **six-month period**. This timeframe provides a sufficient number of observations to estimate reliable portfolio statistics and volatility estimates while minimizing the influence of outdated information that may no longer reflect current market conditions.

0.2 Starting with the Basics: An Equi-balanced Portfolio

All the initial computations and statistical calculations are in the `VaR_Model` worksheet.

0.2.1 Computing Log-Returns

Daily prices were converted into log-returns for all assets. Log-returns are preferred to simple returns due to their mathematically favorable time-aggregation properties, which are particularly important when scaling risk estimates across different time horizons. The log-return is defined as:

$$R_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right)$$

0.2.2 Portfolio Mean and Variance

The portfolio is constructed with equal weights: $w_{DAX} = 0.5$ and $w_{MIB} = 0.5$. Daily portfolio returns are therefore computed as:

$$R_{p,t} = w_{DAX} R_{DAX,t} + w_{MIB} R_{MIB,t}$$

Portfolio variance calculations must account for the correlation between the two indices. Beyond a simple weighted average of individual variances, the covariance term captures the joint movement of the assets:

$$\sigma_p^2 = w_{DAX}^2 \sigma_{DAX}^2 + w_{MIB}^2 \sigma_{MIB}^2 + 2 w_{DAX} w_{MIB} Cov_{DAX,MIB}$$

Implementation Note: *In the provided Excel file, this theoretical property was put into practice by first computing the daily portfolio log-returns row-by-row in a dedicated column, and then directly applying the* `VAR.S()` *and* `AVERAGE()` *functions to that entire array. This computational approach implicitly and perfectly captures the covariance effect described in the formula above.*

These portfolio figures become the foundation for all the risk models that follow. The baseline statistics derived from the 6-month historical window are summarized in Table 1.

Metric	DAX	MIB	Portfolio
Average	-0.04%	0.05%	0.00%
Variance	0.01%	0.01%	0.01%
Sigma (σ)	1.01%	1.05%	0.99%
Sigma EWMA	1.41%	1.41%	1.39%

Table 1: Historical daily averages and volatilities.

The standard portfolio volatility (0.99%) is notably lower than the volatility of either individual index (1.01% and 1.05%). This early observation provides empirical evidence of the diversification benefit at work, preceding any formal Value at Risk calculations.

0.3 Measuring Risk: Parametric Value at Risk

0.3.1 The Parametric VaR

The parametric method assumes returns follow a normal distribution. Given the portfolio mean and volatility, the Value at Risk (maximum plausible loss) for one day can be computed as:

$$VaR_{\text{1-day}}^{1-\alpha} = -\left(\mu_p + z_\alpha \cdot \sigma_p\right)$$

where z_α is the critical value from the normal distribution at chosen confidence levels: 95% ($z = -1.645$), 99% ($z = -2.326$), and 99.5% ($z = -2.576$).

Implementation Note: *In the Excel dashboard, the Parametric VaR is dynamically computed using the* `NORM.S.INV()` *function to extract the exact z_α critical values for the 95%, 99%, and 99.5% probabilities. These values are then multiplied by the portfolio's standard deviation (Sigma) and scaled by the square root of the T horizon specified in the control panel.*

0.3.2 Scaling Over Time

To assess risk over longer periods, the "Square Root of Time" rule is employed. Under the standard assumption of independent daily returns, volatility and VaR scale with $\sqrt{T}$:

$$VaR_T = VaR_{\text{1-day}} \cdot \sqrt{T}$$

Value at Risk estimates were computed for each index individually, their unweighted sum (as a hypothetical undiversified benchmark), and the actual portfolio VaR up to a 100-day horizon.

Implementation Note: *In the Excel* `VaR_Model` *worksheet, this scaling was modeled by creating a sequential timeline from $T = 1$ to $T = 100$. The baseline 1-day Parametric VaR was anchored using absolute cell references (e.g.,* `$J$3`*) and multiplied by the* `SQRT()` *function applied to the respective day index. Dragging this formula down populated the entire dataset required to plot the risk evolution curves.*

You can see how the risk grows in the plot below.

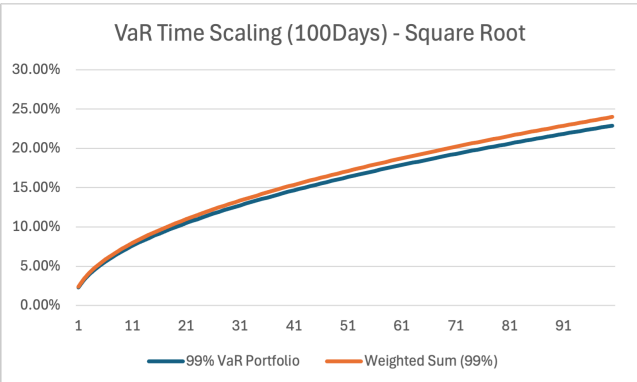

Figure 1: Parametric VaR Scaling over 100 days: Portfolio vs. Undiversified Weighted Sum.

The specific values for the 10-day Parametric VaR, which serves as our baseline benchmark, are reported in Table 2.

Confidence Level	DAX	MIB	Portfolio
95%	5.39%	5.34%	5.12%
99%	7.56%	7.61%	7.24%
99.5%	8.36%	8.44%	8.02%

Table 2: 10-Day Parametric VaR.

These results are consistent with the observed daily volatility statistics. At the 99% confidence level over a 10-day horizon, the model estimates a maximum portfolio loss of 7.24%.

0.4 Adapting to Change: The EWMA Model

A key limitation of the parametric approach is its assumption of constant volatility. In practice, financial markets exhibit *volatility clustering*—periods of relative calm alternating with periods of heightened turbulence. The **EWMA** (Exponentially Weighted Moving Average) model addresses this limitation.

The EWMA methodology assigns greater weight to recent observations while progressively diminishing the influence of older data. The recursive formula is:

$$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)r_{t-1}^2$$

Following the RiskMetrics standard, $\lambda = 0.94$ was employed throughout.

This approach ensures that significant market movements are promptly reflected in updated volatility estimates. The same VaR formula is then applied, incorporating this adaptive volatility measure:

$$VaR_{EWMA} = -\left(\mu + z_\alpha \cdot \sigma_{EWMA}\right) \cdot \sqrt{T}$$

Implementation Note: *In the Excel model, the EWMA volatility was calculated recursively in dedicated columns within the* `VaR_Model` *sheet. Starting from the unconditional historical variance as a seed, the formula iteratively updated the variance day-by-day using the specific daily log-returns. The final, most recent volatility value of this array was then extracted to compute the dynamic EWMA VaR in the Dashboard.*

The resulting risk measure exhibits substantially greater responsiveness to prevailing market conditions. This dynamism becomes apparent upon comparing the 10-day EWMA results (Table 3) with the static Parametric estimates.

Confidence Level	DAX	MIB	Portfolio
95%	7.47%	7.21%	7.21%
99%	10.52%	10.26%	10.20%
99.5%	11.63%	11.37%	11.29%

Table 3: 10-Day EWMA VaR ($\lambda = 0.94$).

The EWMA VaR at 99% for the portfolio reaches 10.20%, notably higher than the 7.24% from the static Parametric model. This divergence reflects a substantive observation: recent trading periods exhibited heightened volatility relative to the 6-month historical mean. The EWMA model appropriately captures this temporal variation, delivering a more conservative risk estimate that reflects current market conditions.

0.5 Simulating the Future: Monte Carlo

To complement the analytical approaches above, a Monte Carlo simulation framework was developed. Rather than relying on closed-form mathematical expressions, this method generates numerous plausible future scenarios to examine the empirical distribution of portfolio outcomes.

The simulation encompassed 10,000 iterations to ensure robust statistical properties. In each iteration, random returns were generated for both assets from normal distributions parameterized by historical mean and volatility estimates, and the corresponding portfolio return was computed.

The 1-day Monte Carlo VaR is obtained by sorting the 10,000 simulated portfolio returns from worst to best and identifying the value at the $(1-\alpha)$ percentile:

$$VaR_{\text{1-day}}^{MC} = -\text{Percentile}\left(\{R_{sim,i}\}_{i=1}^{N}, 1-\alpha\right)$$

Implementation Note: *In the Excel framework, this stochastic process was executed in the* `VaR_Model` *sheet by generating an array of 10,000 random normal scenarios for both assets using inverse normal transformation functions coupled with random probability generators (e.g.,* `NORM.INV(RAND(), ...)`*). The portfolio simulated returns were aggregated in a dedicated column, and the VaR was extracted dynamically using the* `PERCENTILE.INC()` *function, capturing the exact tail risk from the empirical simulated distribution.*

The 1-day results are subsequently scaled to a 10-day horizon using the square root of time principle:

$$VaR_{T\text{-day}}^{MC} = VaR_{\text{1-day}}^{MC} \cdot \sqrt{10}$$

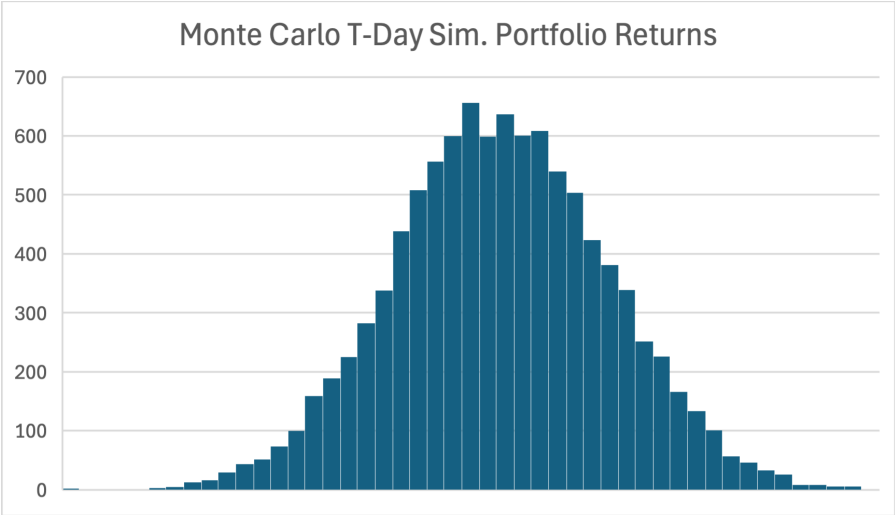

Figure 2: Distribution of Simulated Portfolio Returns (N = 10,000 iterations).

Table 4 presents the output of a specific simulation run.

Confidence Level	DAX	MIB	Portfolio
95%	5.69%	5.07%	5.14%
99%	7.84%	7.35%	7.34%
99.5%	8.52%	8.05%	8.03%

Table 4: 10-Day Monte Carlo VaR (N = 10,000).

It is important to note that due to the stochastic nature of the Monte Carlo method, these exact values will marginally fluctuate upon every recalculation of the model. However, thanks to the Law of Large Numbers (ensured by the massive N=10,000 iterations), the simulated Portfolio VaR at 99% (7.34%) firmly converges around the theoretical Parametric VaR (7.24%), confirming the structural validity of the generation engine.

0.6 Learning from the Past: Historical VaR

The Historical method represents the most straightforward approach, relying on empirical data without imposing distributional assumptions. This non-parametric technique allows observed historical outcomes to directly inform risk assessment.

0.6.1 1-Day Historical VaR

Actual historical returns across the six-month sample period are sorted from worst to best performance. The Value at Risk is then determined as the value at the $(1-\alpha)$ percentile of this empirical distribution:

$$VaR^{Hist}_{\text{1-day}} = -\text{Percentile}\left(\{R_{historical}\}, 1-\alpha\right)$$

0.6.2 T-Day Historical VaR

I scale the 1-day historical values to a 10-day horizon using the same logic. It's a useful comparison to see how the other methods stack up against actual historical data.

Table 5 summarizes both the 1-day Historical VaR and its 10-day scaled projection.

	1-Day Historical VaR			10-Day Scaled VaR		
Conf. Level	DAX	MIB	Portfolio	DAX	MIB	Portfolio
95%	1.74%	1.75%	1.63%	5.50%	5.54%	5.16%
99%	2.80%	2.31%	2.54%	8.86%	7.29%	8.02%
99.5%	3.11%	2.99%	3.05%	9.83%	9.46%	9.65%

Table 5: Historical VaR estimates (1-Day and 10-Day scaled).

Notably, the 10-day Historical Simulation VaR for the portfolio at 99% (8.02%) exceeds the Parametric estimate (7.24%). This difference underscores a fundamental advantage of the historical methodology: it does not impose a normal distribution assumption and thereby naturally incorporates "fat tails" (the empirically observed frequency of extreme events over the past six months), a phenomenon that parametric approaches tend to underestimate.

0.7 Testing the Value of Diversification

A key property of a good risk measure is **subadditivity**: the risk of a diversified portfolio must be less than or equal to the sum of individual component risks:

$$VaR_\alpha(w_1 A + w_2 B) \leq w_1 VaR_\alpha(A) + w_2 VaR_\alpha(B)$$

I tested this across all four models (Parametric, EWMA, Monte Carlo, and Historical) at the 99% confidence level for 10 days. In every case, the actual portfolio VaR is lower than the simple weighted sum of individual index risks.

To validate this property empirically, the 99% VaR at 10 days was computed across all four methodologies and the actual Portfolio VaR was compared against the simple weighted sum of component risks (Table 6).

Implementation Note: *In the final Dashboard, this theoretical axiom is tested dynamically using an* `IF()` *logical statement. The spreadsheet calculates the linear weighted sum of the standalone VaRs and compares it to the portfolio VaR. The condition* `IF(Actual < Weighted, "YES", "NO")` *strictly validates the presence of the diversification benefit. Conditional formatting is applied to the output cells to provide immediate visual confirmation of the subadditivity.*

Model	Weighted Sum	Actual Portfolio	Subadditive?
Parametric	7.59%	7.24%	**YES**
EWMA	10.39%	10.20%	**YES**
Monte Carlo	7.60%	7.34%	**YES**
Historical Sim	8.07%	8.02%	**YES**

Table 6: Subadditivity Check across all models (99% Confidence, T=10).

The empirical evidence is absolute: regardless of the underlying mathematical assumptions—whether static (Parametric), dynamic (EWMA), stochastic (Monte Carlo), or empirical (Historical)—the actual portfolio VaR is systematically lower than the weighted sum of individual index risks.

This is the tangible value of the **diversification benefit**. Because DAX and FTSE MIB are not perfectly correlated ($\rho < 1$), combining them structurally dampens the overall exposure. This benefit is also visually confirmed by the gap in the scaling plot (Figure 1), where the true portfolio risk trajectory sits safely below the theoretical undiversified boundary.